

Vo Quoc Huy

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EXPERIENCE

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Ho Chi Minh City, Vietnam

Software Engineer (FastAPI, Azure, Terraform, MySQL, Kubernetes, AI Agents)

May 2025 – Present

- Architected and delivered a distributed workflow platform for engineering document synchronization, replacing scheduled batch execution with a fault-tolerant event-driven architecture that reduced processing latency from 15 minutes to under 3 minutes.
- Architected and delivered a GitOps platform for Kubernetes using ArgoCD, enabling declarative infrastructure, automatic reconciliation, and reducing production recovery time from hours to minutes.
- Developed a proof-of-concept CDC pipeline (MySQL → Kafka → ClickHouse) to evaluate OLAP adoption, reducing representative analytical queries from 6 hours to 30 seconds while evaluating production tradeoffs.
- Built a document mapping system using Azure Document Intelligence and custom change-detection logic to extract structured data from high-volume PDF processing workloads, significantly reducing manual processing effort.
- Built a high-performance reporting engine using asynchronous pipelines and precomputation, reducing query latency and cloud costs by ~50% for internal analytics workloads.
- Built an internal platform exposing backend services via OpenAPI contracts, enabling self-service workflow orchestration for frontend teams and AI agents; integrated AI-driven automation into CI/CD pipelines, accelerating release velocity by ~2-3× while eliminating third-party vendor costs.
- Established standardized service-to-service communication for a microservices platform, evaluating gRPC and private REST tradeoffs while implementing shared authentication, API contracts, and observability.

Employment Hero

Remote

Backend Software Engineer (Ruby on Rails, PostgreSQL, gRPC, CICD)

Apr 2024 – Apr 2025

- Designed a scalable search system using PostgreSQL (tsvector, trigram), providing a cost-effective alternative to expensive external search infrastructure.
- Identified and resolved a critical authentication vulnerability related to token misuse; collaborated with team leads to implement session-handling safeguards.
- Designed and integrated Braze webhooks to automate WhatsApp notifications for shortlisted candidates in Malaysia–Singapore, improving communication efficiency and reducing manual intervention.
- Contributed to a 5x cost reduction in CI/CD by migrating all service pipelines from CircleCI to GitHub Actions.
- Optimized gRPC communication and complex database queries to resolve critical performance bottlenecks.
- Built a cross-device auto-save mechanism using Redis caching that directly improved platform retention metrics.

Athena Studio

Ho Chi Minh City, Vietnam

Software Engineer (Python, FastAPI, GCP, Airflow)

Nov 2022 – Mar 2024

- Developed high-throughput ingestion pipelines using FastAPI for automated revenue and cost data collection at scale.
- Implemented event-driven processing systems via GCP Pub/Sub, handling millions of daily events while cutting infra costs by ~25%.
- Contributed to a strategic refactor of legacy systems and updated internal dashboards using Airflow, FastAPI, improving feature delivery efficiency by ~35%.

EDUCATION

Royal Melbourne Institute of Technology Vietnam

Ho Chi Minh City, Vietnam

Bachelor of Software Engineering (First Class Honours)

Sep 2019 – Sep 2023

SKILLS

Systems & Architecture: Distributed Systems, Event-Driven Architecture, Microservices, Scalable System Design.

Backend Development: Python (Django, FastAPI), Ruby on Rails, TypeScript, API Design (REST, gRPC, GraphQL).

Frontend: ReactJS (working knowledge).

Data & Streaming: PostgreSQL, Redis, Kafka, SQL Optimization, Data Pipelines, Airflow.

Cloud & DevOps: Terraform, Azure, GCP, Docker, Kubernetes, CI/CD (GitHub Actions, Azure DevOps).

AI Engineering: LLM Integration, RAG Architectures, AI Workflow Integration.

Languages: English (Full professional proficiency), Vietnamese (Native).